Enterprise Zero Trust Deployment Lab

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Tenant: CorporationABC.onmicrosoft.com

Microsoft Intune & Entra ID Security Deployment

This project demonstrates the implementation of Zero Trust principles by securing identities, enforcing access controls, and managing endpoints using Microsoft Entra ID and Intune.

Key Focus Areas

- Identity Protection (MFA & Conditional Access)
- Endpoint Security (Compliance & Hardening)
- Role-Based Access Control
- Device Management via Intune

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Tenant: CorporationABC.onmicrosoft.com
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Architecture Overview

- Identity: Microsoft Entra ID
- Device Management: Microsoft Intune
- Security Controls: Conditional Access + MFA
- Endpoint Protection: Microsoft Defender (planned/partial)

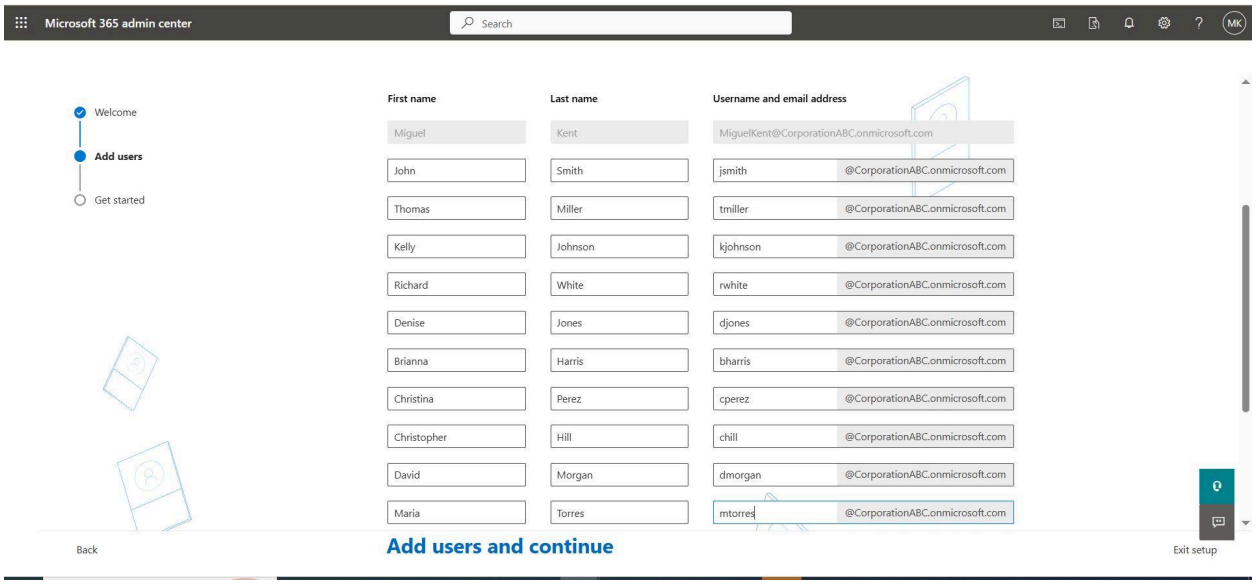
User Configuration

Description:
Created 10 user accounts to simulate a small business environment within the tenant.

Details:

- Tenant: CorporationABC.onmicrosoft.com
- Total Users Created: 10

Purpose:
To simulate a real-world organization with multiple users for identity and access management testing.



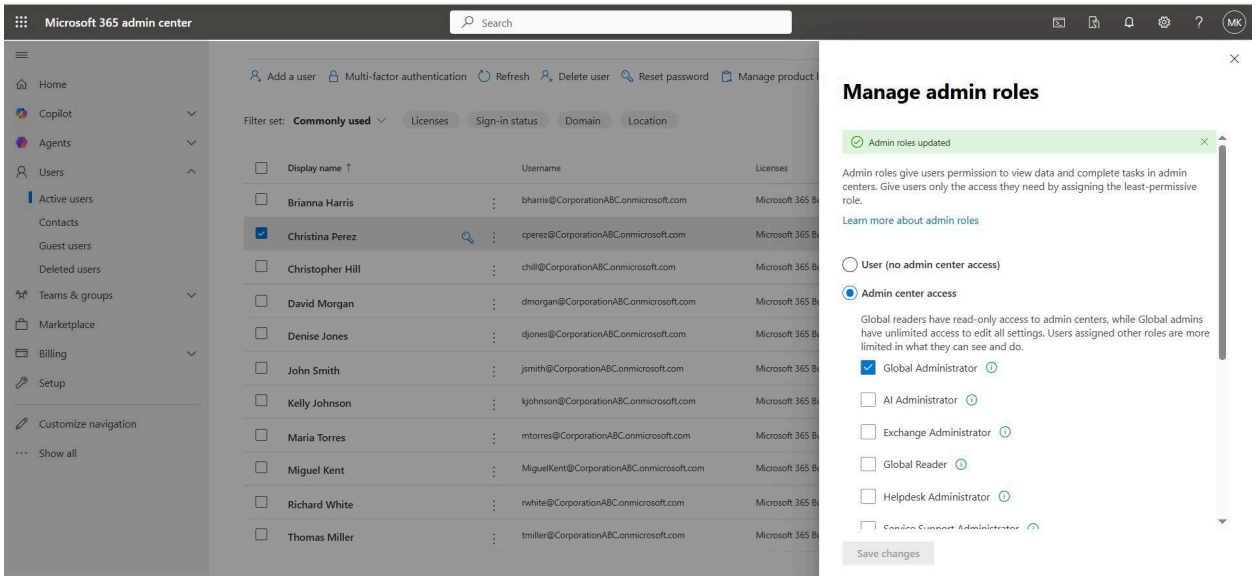
Role Assignment

Description:
Configured administrative access within the tenant.

Configuration:

- Global Administrator: cperez@CorporationABC.onmicrosoft.com
- All other users assigned as standard users

Purpose:
To maintain centralized administrative control while limiting elevated privileges.



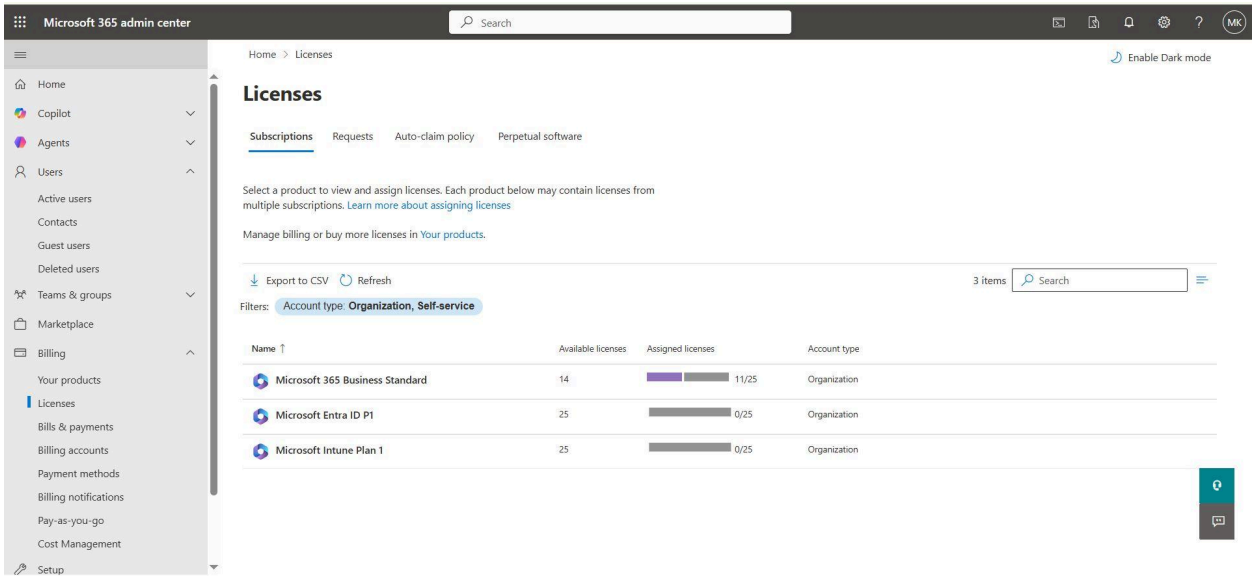
Licensing Configuration

Description:
Configured licensing to enable identity protection and device management features.

Configuration:

- Microsoft Entra ID Plan 1 enabled
- Microsoft Intune Plan 1 enabled

Purpose:
To support Conditional Access and endpoint security capabilities.



Multi-Factor Authentication (Conditional Access)

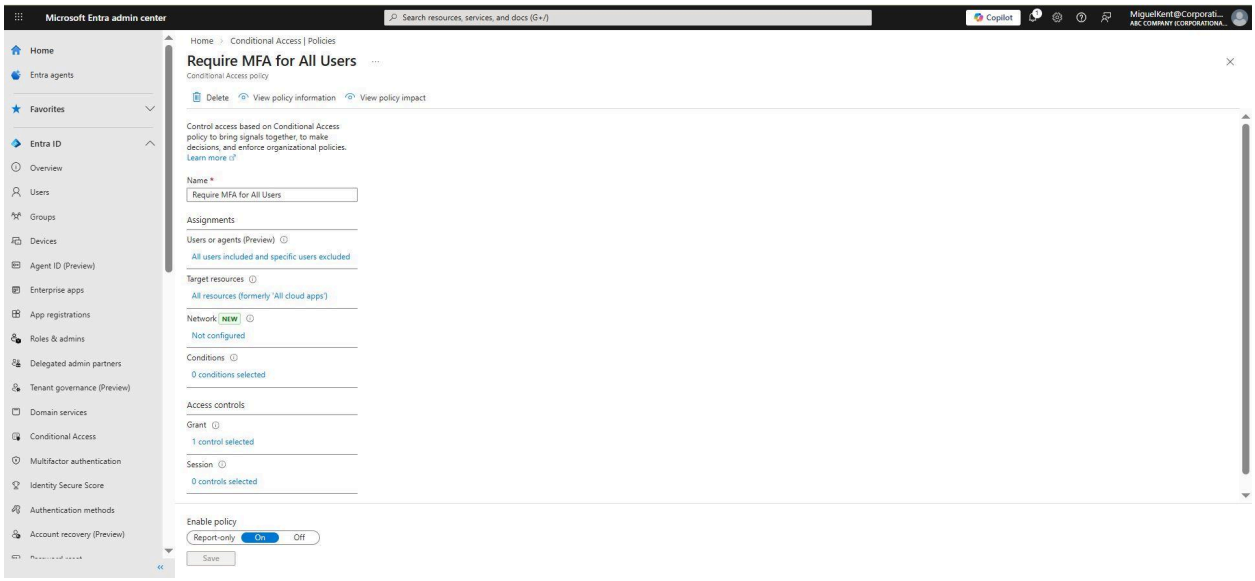
Description:
Implemented a Conditional Access policy to enforce multi-factor authentication across all users.

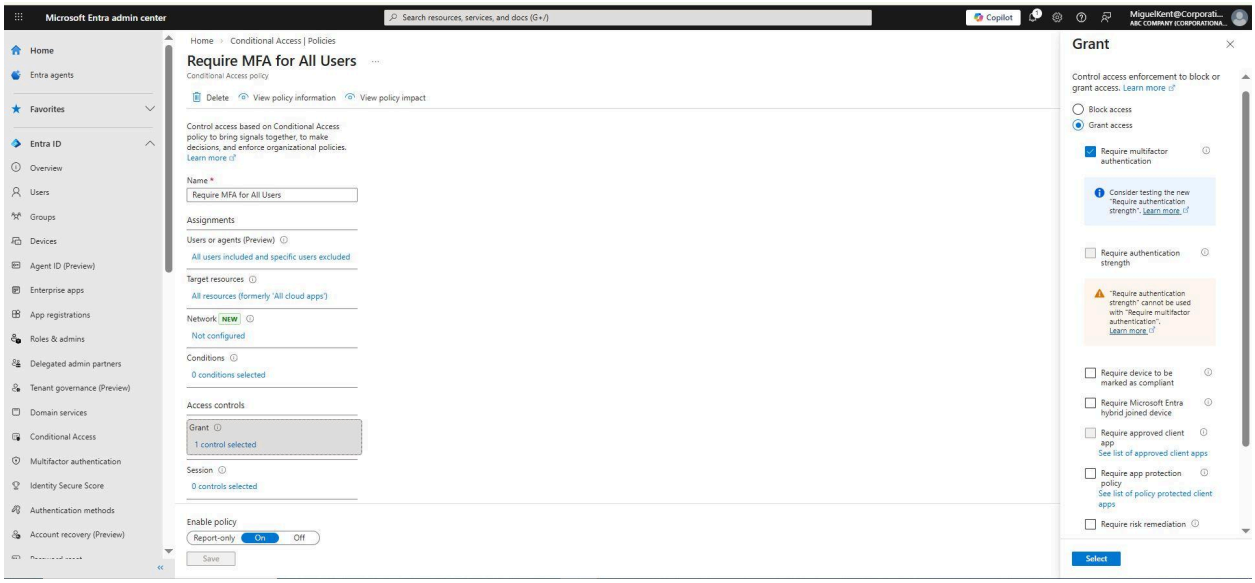
Configuration:

- Applied to all users
- Administrator account excluded to prevent lockout
- Targeted all cloud applications
- Required MFA for access

Purpose:
To protect against unauthorized access and credential-based attacks.

Result:
All users must complete multi-factor authentication when accessing organizational resources.





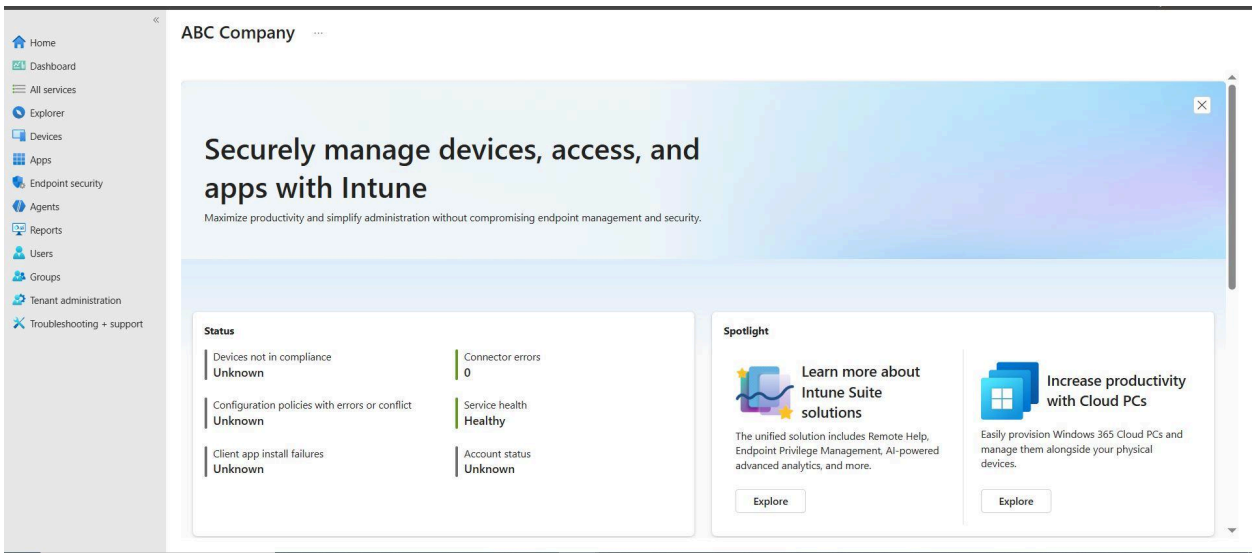
Intune Initialization

Description:
Accessed Microsoft Intune to begin device management configuration.

Environment:

- Platform: Microsoft Intune
- Tenant: CorporationABC.onmicrosoft.com

Purpose:
To manage and secure endpoint devices through centralized policies.



Device Enrollment Configuration

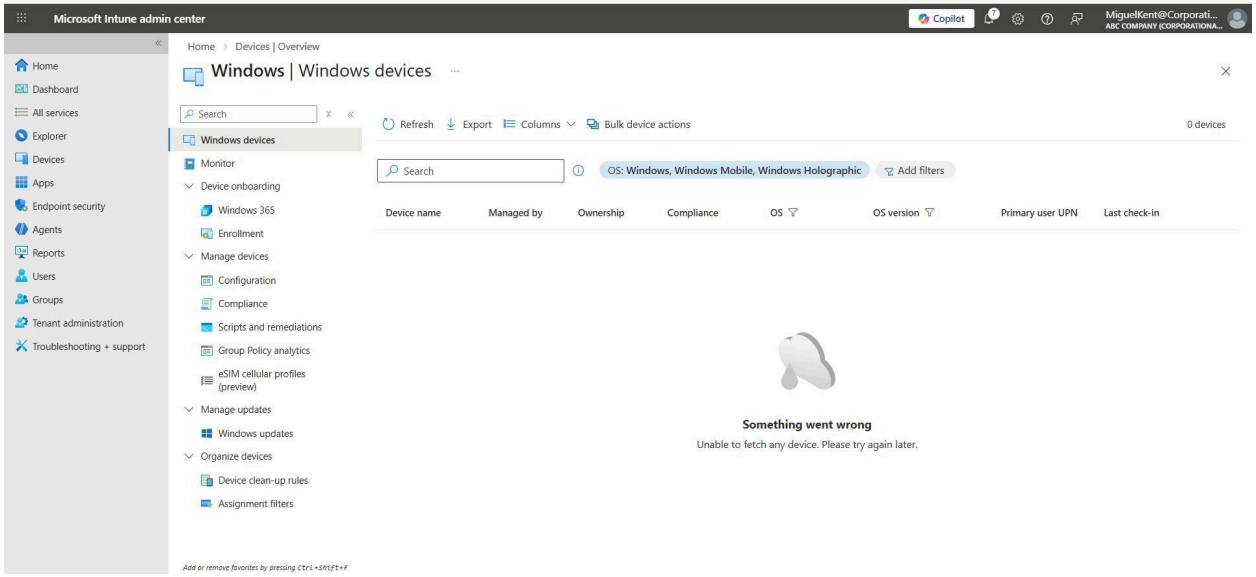
Description:
Configured the Intune environment for device management.

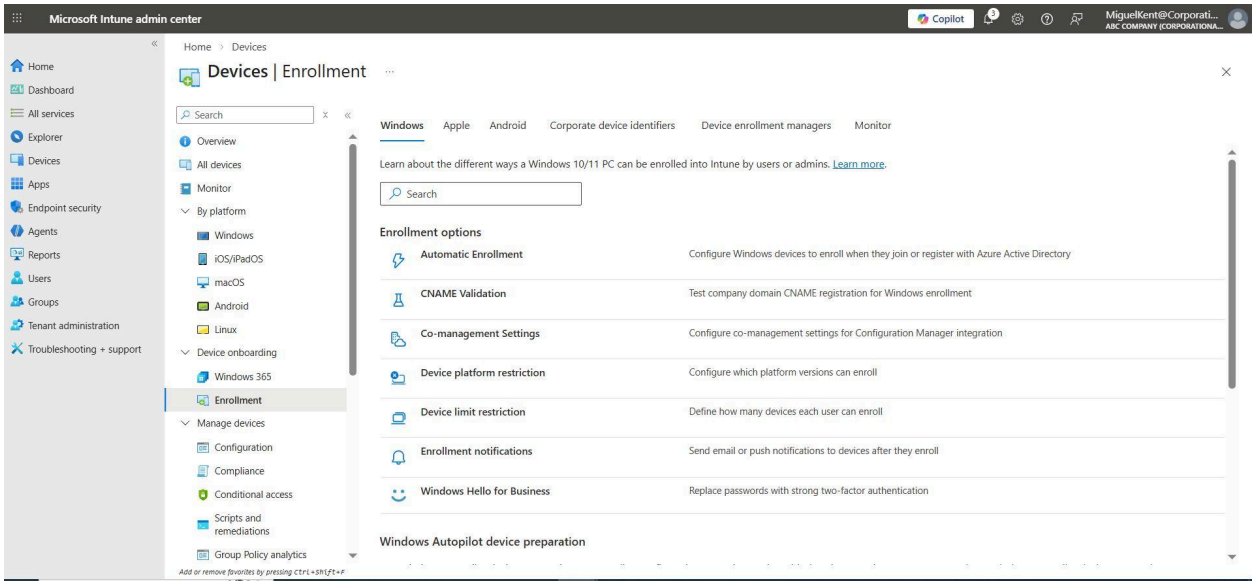
Location:
Devices → Windows → Windows devices

Configuration:

- Accessed Windows device management section
- Prepared environment for device onboarding

Purpose:
To establish a foundation for managing endpoint devices.





Device Compliance Policy

Description:
Created a Windows compliance policy to enforce security requirements.

Configuration:

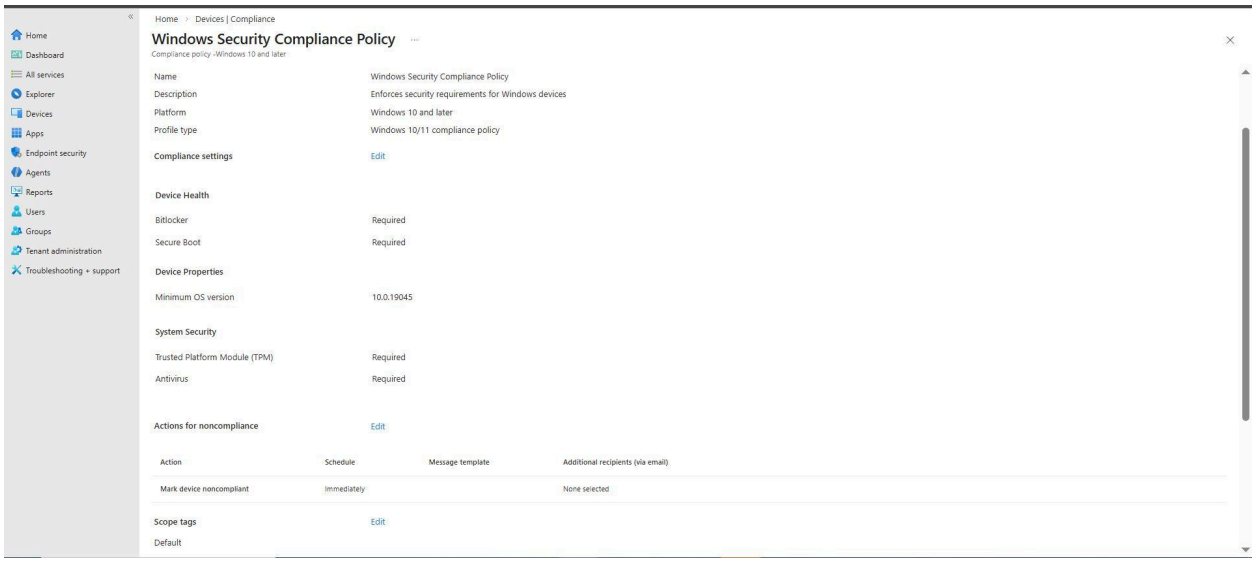
- BitLocker required

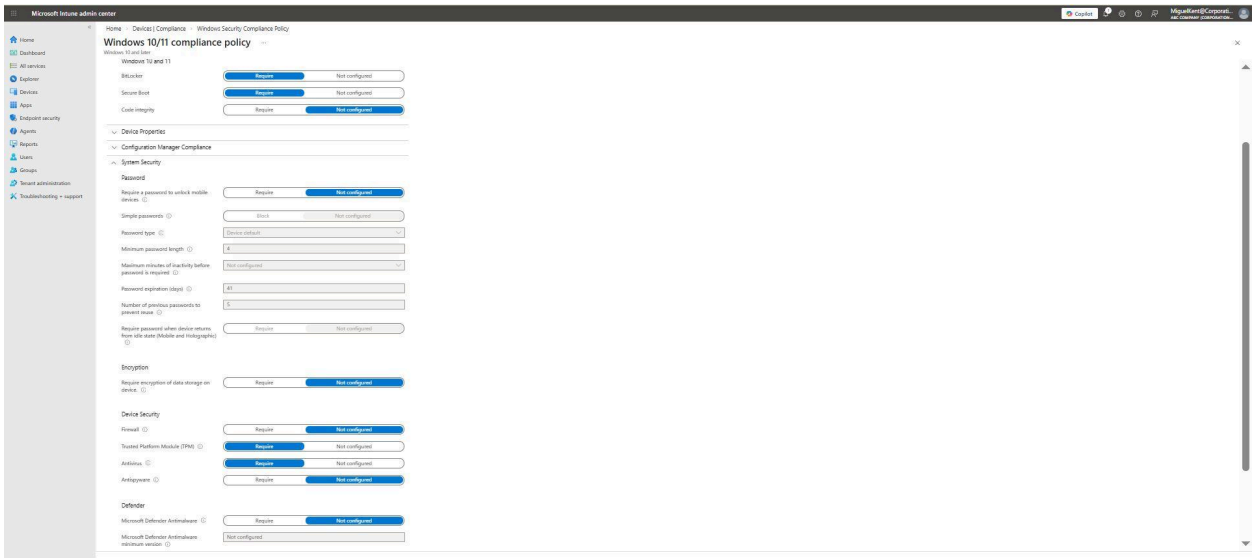
- Secure Boot required
- TPM required
- Antivirus required
- Minimum OS version enforced

Assignment:

- Applied to all users

Purpose:
To ensure devices meet security standards before accessing resources.





Configuration Policy (OS Hardening)

Description:
Created a configuration policy to enforce security settings on Windows devices.

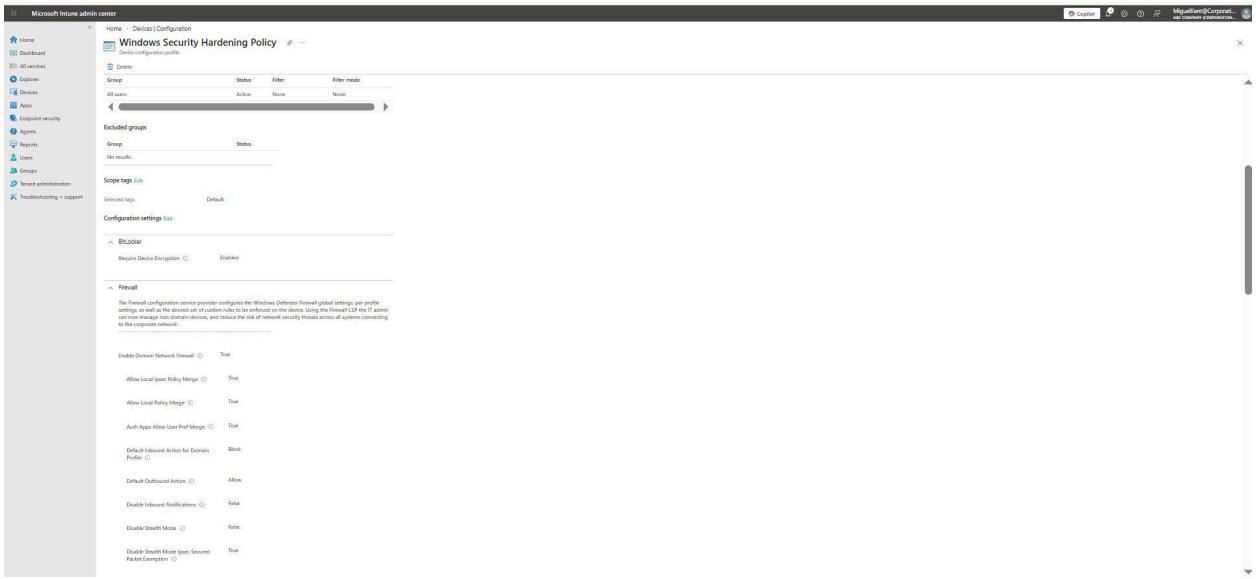
Configuration:

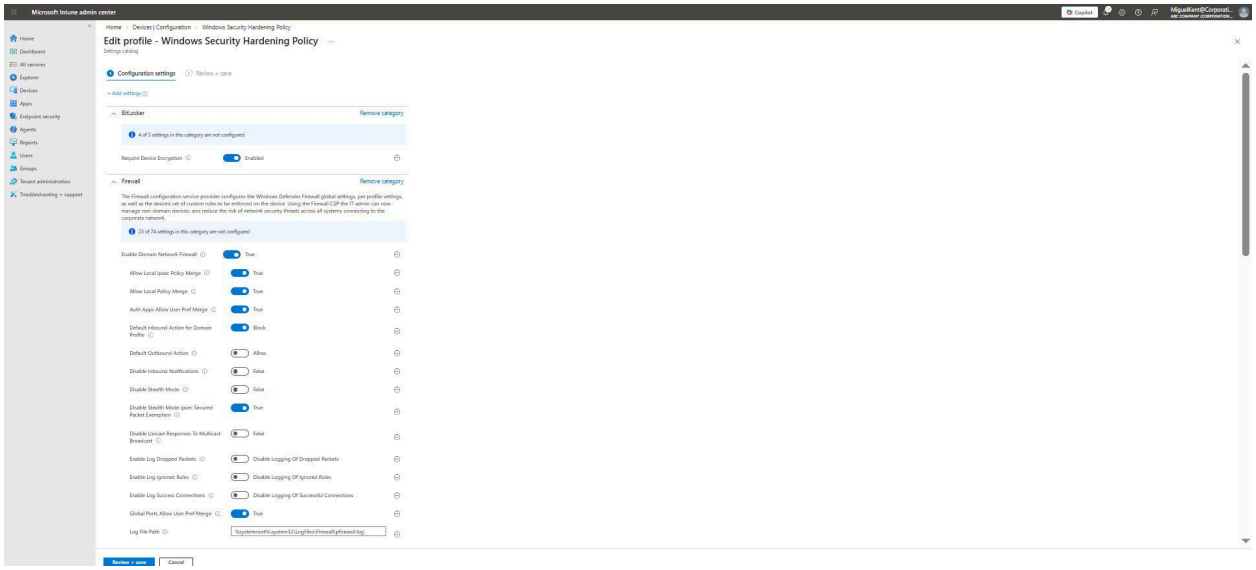
- BitLocker encryption enabled
- Windows Firewall enabled (Domain, Private, Public)

Assignment:

- Applied to all users

Purpose:
To harden endpoint devices and enforce baseline security configurations





Zero Trust Architecture Overview

Users
↓
Microsoft Entra ID (Identity)
↓
Conditional Access (MFA Enforcement)
↓
Microsoft Intune (Device Management)
↓
Compliance Policies (Device Validation)
↓
Configuration Policies (OS Hardening)
↓
Secure Access to Microsoft 365 Resources

This architecture follows a Zero Trust model where identity is continuously verified, devices must meet compliance standards, and access is controlled through policy enforcement.